

ORDERS TABLE _SQL AGGREGATE & CLAUSE OPERATIONS

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Question :

Perform SQL operations on an Orders table using WHERE, ORDER BY, GROUP BY, HAVING, SUM(), AVG(), MAX(), MIN(), COUNT(), and BETWEEN to analyse orders, customer purchases, salesman performance, and purchase amounts.

Step 1 : Create Database and Table

```
1  -- Step 1: Create Database and Table
2  • CREATE DATABASE Ordersdb;
3  • USE Ordersdb;
4  • CREATE TABLE orders (
5      ord_no INT PRIMARY KEY,
6      purch_amt DECIMAL(10,2),
7      ord_date DATE,
8      customer_id INT,
9      salesman_id INT
10 );
```

Output

#	Time	Action	Message
✓ 1	23:22:23	CREATE DATABASE Ordersdb	1 row(s) affected
✓ 2	23:22:41	USE Ordersdb	0 row(s) affected
✓ 3	23:22:44	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, cus...	0 row(s) affected

```

11  -- Step 2: Insert Data
12  • INSERT INTO orders VALUES
13    (70001,150.50,'2012-10-05',3005,5002),
14    (70009,270.65,'2012-09-10',3001,5005),
15    (70002,65.26,'2012-10-05',3002,5001),
16    (70004,110.50,'2012-08-17',3009,5003),
17    (70007,948.50,'2012-09-10',3005,5002),
18    (70005,2400.60,'2012-07-27',3007,5001),
19    (70008,5760.00,'2012-09-10',3002,5001),
20    (70010,1983.43,'2012-10-10',3004,5006),
21    (70003,2480.40,'2012-10-10',3009,5003),
22    (70012,250.45,'2012-06-27',3008,5002),
23    (70011,75.29,'2012-08-17',3003,5007),
24    (70013,3045.60,'2012-04-25',3002,5001);

```

Output

Action Output

#	Time	Action	Message
✓ 1	23:22:23	CREATE DATABASE Ordersdb	1 row(s) affected
✓ 2	23:22:41	USE Ordersdb	0 row(s) affected
✓ 3	23:22:44	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, cus...	0 row(s) affected
✓ 4	23:24:10	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005), ...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0

Step 2 : WHERE

```

25  -- Step 3: WHERE Clause
26  • SELECT *
27    FROM orders
28    WHERE purch_amt > 2000;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
*	NULL	NULL	NULL	NULL	NULL

orders 1 x

Output

Action Output

#	Time	Action	Message
✓ 1	23:22:23	CREATE DATABASE Ordersdb	1 row(s) affected
✓ 2	23:22:41	USE Ordersdb	0 row(s) affected
✓ 3	23:22:44	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, cu...	0 row(s) affected
✓ 4	23:24:10	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005)...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
✓ 5	23:24:56	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000	4 row(s) returned

```

29 -- Step 4: WHERE Clause with Date
30 • SELECT *
31 FROM orders
32 WHERE ord_date = '2012-09-10';

```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70007	948.50	2012-09-10	3005	5002
70008	5760.00	2012-09-10	3002	5001
70009	270.65	2012-09-10	3001	5005
NULL	NULL	NULL	NULL	NULL

orders 2 x

#	Time	Action	Message
1	23:22:23	CREATE DATABASE Ordersdb	1 row(s) affected
2	23:22:41	USE Ordersdb	0 row(s) affected
3	23:22:44	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, cu...	0 row(s) affected
4	23:24:10	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,5005)...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
5	23:24:56	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000	4 row(s) returned
6	23:25:39	SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1000	3 row(s) returned

```

33 -- Step 5: WHERE Clause with Salesman
34 • SELECT *
35 FROM orders
36 WHERE salesman_id = 5001;

```

ord_no	purch_amt	ord_date	customer_id	salesman_id
70002	65.26	2012-10-05	3002	5001
70005	2400.60	2012-07-27	3007	5001
70008	5760.00	2012-09-10	3002	5001
70013	3045.60	2012-04-25	3002	5001
NULL	NULL	NULL	NULL	NULL

orders 3 x

#	Time	Action	Message
2	23:22:41	USE Ordersdb	0 row(s) affected
3	23:22:44	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, ...	0 row(s) affected
4	23:24:10	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,500...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
5	23:24:56	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000	4 row(s) returned
6	23:25:39	SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1000	3 row(s) returned
7	23:26:10	SELECT * FROM orders WHERE salesman_id = 5001 LIMIT 0, 1000	4 row(s) returned

Step 3 : ORDER BY

```

37  -- Step 6: ORDER BY Purchase Amount
38  • SELECT *
39    FROM orders
40    ORDER BY purch_amt DESC;

```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70008	5760.00	2012-09-10	3002	5001
	70013	3045.60	2012-04-25	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70005	2400.60	2012-07-27	3007	5001
	70010	1983.43	2012-10-10	3004	5006
	70007	948.50	2012-09-10	3005	5002
	70009	270.65	2012-09-10	3001	5005
	70012	250.45	2012-06-27	3008	5002
	70001	150.50	2012-10-05	3005	5002
	70004	110.50	2012-08-17	3009	5003
	70011	75.29	2012-08-17	3003	5007
	70002	65.26	2012-10-05	3002	5001
•	NULL	NULL	NULL	NULL	NULL

orders 4 x

Output

Action Output

#	Time	Action	Message
3	23:22:44	CREATE TABLE orders (ord_no INT PRIMARY KEY, purch_amt DECIMAL(10,2), ord_date DATE, ...	0 row(s) affected
4	23:24:10	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,500...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
5	23:24:56	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000	4 row(s) returned
6	23:25:39	SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1000	3 row(s) returned
7	23:26:10	SELECT * FROM orders WHERE salesman_id = 5001 LIMIT 0, 1000	4 row(s) returned
8	23:27:09	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000	12 row(s) returned

```

41  -- Step 7: ORDER BY Order Date
42  • SELECT *
43    FROM orders
44    ORDER BY ord_date ASC;

```

Result Grid

Filter Rows:

Edit: Export/Import: Wrap Cell Content:

	ord_no	purch_amt	ord_date	customer_id	salesman_id
▶	70013	3045.60	2012-04-25	3002	5001
	70012	250.45	2012-06-27	3008	5002
	70005	2400.60	2012-07-27	3007	5001
	70004	110.50	2012-08-17	3009	5003
	70011	75.29	2012-08-17	3003	5007
	70007	948.50	2012-09-10	3005	5002
	70008	5760.00	2012-09-10	3002	5001
	70009	270.65	2012-09-10	3001	5005
	70001	150.50	2012-10-05	3005	5002
	70002	65.26	2012-10-05	3002	5001
	70003	2480.40	2012-10-10	3009	5003
	70010	1983.43	2012-10-10	3004	5006
•	NULL	NULL	NULL	NULL	NULL

orders 5 x

Output

Action Output

#	Time	Action	Message
4	23:24:10	INSERT INTO orders VALUES (70001,150.50,'2012-10-05',3005,5002), (70009,270.65,'2012-09-10',3001,500...	12 row(s) affected Records: 12 Duplicates: 0 Warnings: 0
5	23:24:56	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000	4 row(s) returned
6	23:25:39	SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1000	3 row(s) returned
7	23:26:10	SELECT * FROM orders WHERE salesman_id = 5001 LIMIT 0, 1000	4 row(s) returned
8	23:27:09	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000	12 row(s) returned
9	23:27:41	SELECT * FROM orders ORDER BY ord_date ASC LIMIT 0, 1000	12 row(s) returned

Step 4 : AGGREGATE FUNCTIONS

```

45  -- Step 8: SUM()
46  • SELECT SUM(purch_amt) AS total_revenue
47  FROM orders;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
total_revenue			
17541.18			

Result 6 x

Output

Action Output

#	Time	Action	Message
✓ 5	23:24:56	SELECT * FROM orders WHERE purch_amt > 2000 LIMIT 0, 1000	4 row(s) returned
✓ 6	23:25:39	SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1000	3 row(s) returned
✓ 7	23:26:10	SELECT * FROM orders WHERE salesman_id = 5001 LIMIT 0, 1000	4 row(s) returned
✓ 8	23:27:09	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000	12 row(s) returned
✓ 9	23:27:41	SELECT * FROM orders ORDER BY ord_date ASC LIMIT 0, 1000	12 row(s) returned
✓ 10	23:28:41	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000	1 row(s) returned

```

48  -- Step 9: AVG()
49  • SELECT AVG(purch_amt) AS average_order
50  FROM orders;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
average_order			
1461.765000			

Result 7 x

Output

Action Output

#	Time	Action	Message
✓ 6	23:25:39	SELECT * FROM orders WHERE ord_date = '2012-09-10' LIMIT 0, 1000	3 row(s) returned
✓ 7	23:26:10	SELECT * FROM orders WHERE salesman_id = 5001 LIMIT 0, 1000	4 row(s) returned
✓ 8	23:27:09	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000	12 row(s) returned
✓ 9	23:27:41	SELECT * FROM orders ORDER BY ord_date ASC LIMIT 0, 1000	12 row(s) returned
✓ 10	23:28:41	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 11	23:29:10	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000	1 row(s) returned

```

51 -- Step 10: MAX()
52 • SELECT MAX(purch_amt) AS highest_order
53 FROM orders;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
highest_order			
▶ 5760.00			

Result 9 x

Output

Action Output

#	Time	Action	Message
✓ 8	23:27:09	SELECT * FROM orders ORDER BY purch_amt DESC LIMIT 0, 1000	12 row(s) returned
✓ 9	23:27:41	SELECT * FROM orders ORDER BY ord_date ASC LIMIT 0, 1000	12 row(s) returned
✓ 10	23:28:41	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 11	23:29:10	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 12	23:29:49	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 13	23:30:48	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned

```

54 -- Step 11: MIN()
55 • SELECT MIN(purch_amt) AS lowest_order
56 FROM orders;

```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
lowest_order			
▶ 65.26			

Result 10 x

Output

Action Output

#	Time	Action	Message
✓ 9	23:27:41	SELECT * FROM orders ORDER BY ord_date ASC LIMIT 0, 1000	12 row(s) returned
✓ 10	23:28:41	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 11	23:29:10	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 12	23:29:49	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 13	23:30:48	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 14	23:31:32	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000	1 row(s) returned

```

57 -- Step 12: COUNT()
58 • SELECT COUNT(*) AS total_orders
59 FROM orders;

```

total_orders
12

Result 11 x

Output

Action Output

#	Time	Action	Message
✓ 10	23:28:41	SELECT SUM(purch_amt) AS total_revenue FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 11	23:29:10	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 12	23:29:49	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 13	23:30:48	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 14	23:31:32	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 15	23:32:05	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000	1 row(s) returned

Step 5 : GROUP BY

```

60 -- Step 13: GROUP BY Salesman
61 • SELECT salesman_id,
62        SUM(purch_amt) AS total_sales
63 FROM orders
64 GROUP BY salesman_id;

```

salesman_id	total_sales
5002	1349.45
5001	11271.46
5003	2590.90
5005	270.65
5006	1983.43
5007	75.29

Result 12 x

Output

Action Output

#	Time	Action	Message
✓ 11	23:29:10	SELECT AVG(purch_amt) AS average_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 12	23:29:49	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 13	23:30:48	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 14	23:31:32	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 15	23:32:05	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 16	23:33:30	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1...	6 row(s) returned

```

65 -- Step 14: GROUP BY Customer
66 • SELECT customer_id,
67       SUM(purch_amt) AS total_purchase
68 FROM orders
69 GROUP BY customer_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	total_purchase
▶	3005	1099.00
	3002	8870.86
	3009	2590.90
	3007	2400.60
	3001	270.65
	3004	1983.43
	3003	75.29
	3008	250.45

Result 13 x

Output

Action Output

#	Time	Action	Message
✓	12 23:29:49	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓	13 23:30:48	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓	14 23:31:32	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓	15 23:32:05	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000	1 row(s) returned
✓	16 23:33:30	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1...	6 row(s) returned
✓	17 23:34:06	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT ...	8 row(s) returned

```

70 -- Step 15: GROUP BY + MAX()
71 • SELECT customer_id,
72       MAX(purch_amt) AS highest_purchase
73 FROM orders
74 GROUP BY customer_id;

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	customer_id	highest_purchase
▶	3005	948.50
	3002	5760.00
	3009	2480.40
	3007	2400.60
	3001	270.65
	3004	1983.43
	3003	75.29
	3008	250.45

Result 14 x

Output

Action Output

#	Time	Action	Message
✓	13 23:30:48	SELECT MAX(purch_amt) AS highest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓	14 23:31:32	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓	15 23:32:05	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000	1 row(s) returned
✓	16 23:33:30	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1...	6 row(s) returned
✓	17 23:34:06	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT ...	8 row(s) returned
✓	18 23:34:46	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIM...	8 row(s) returned

Step 6 : GROUP BY +HAVING


```

75  -- Step 16: GROUP BY + HAVING
76  • SELECT salesman_id,
77        SUM(purch_amt) AS total_sales
78  FROM orders
79  GROUP BY salesman_id
80  HAVING SUM(purch_amt) > 3000;

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	salesman_id	total_sales
▶	5001	11271.46

Result 15 ×

Output

Action Output

#	Time	Action	Message
✓ 14	23:31:32	SELECT MIN(purch_amt) AS lowest_order FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 15	23:32:05	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 16	23:33:30	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1...	6 row(s) returned
✓ 17	23:34:06	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT ...	8 row(s) returned
✓ 18	23:34:46	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIM...	8 row(s) returned
✓ 19	23:36:26	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING S...	1 row(s) returned

```

81  -- Step 17: GROUP BY + HAVING
82  • SELECT customer_id,
83        SUM(purch_amt) AS total_purchase
84  FROM orders
85  GROUP BY customer_id
86  HAVING SUM(purch_amt) > 2500;

```

Result Grid Filter Rows: Export: Wrap Cell Content:

	customer_id	total_purchase
▶	3002	8870.86
	3009	2590.90

Result 16 ×

Output

Action Output

#	Time	Action	Message
✓ 15	23:32:05	SELECT COUNT(*) AS total_orders FROM orders LIMIT 0, 1000	1 row(s) returned
✓ 16	23:33:30	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1...	6 row(s) returned
✓ 17	23:34:06	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT ...	8 row(s) returned
✓ 18	23:34:46	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIM...	8 row(s) returned
✓ 19	23:36:26	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING S...	1 row(s) returned
✓ 20	23:37:13	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	2 row(s) returned

```

87  -- Step 18: GROUP BY + HAVING
88  • SELECT customer_id,
89        COUNT(*) AS total_orders
90  FROM orders
91  GROUP BY customer_id
92  HAVING COUNT(*) > 1;

```

customer_id	total_orders
3005	2
3002	3
3009	2

Result 17 x

Output

#	Time	Action	Message
✓ 16	23:33:30	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id LIMIT 0, 1...	6 row(s) returned
✓ 17	23:34:06	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT ...	8 row(s) returned
✓ 18	23:34:46	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIM...	8 row(s) returned
✓ 19	23:36:26	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING S...	1 row(s) returned
✓ 20	23:37:13	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	2 row(s) returned
✓ 21	23:38:05	SELECT customer_id, COUNT(*) AS total_orders FROM orders GROUP BY customer_id HAVING COUNT...	3 row(s) returned

Step 7: GROUP BY +HAVING+ORDER BY

```

93  -- Step 19: GROUP BY + HAVING + ORDER BY
94  • SELECT customer_id,
95        SUM(purch_amt) AS total_purchase
96  FROM orders
97  GROUP BY customer_id
98  HAVING SUM(purch_amt) > 1000
99  ORDER BY total_purchase DESC;

```

customer_id	total_purchase
3002	8870.86
3009	2590.90
3007	2400.60
3004	1983.43
3005	1099.00

Result 18 x

Output

#	Time	Action	Message
✓ 17	23:34:06	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id LIMIT ...	8 row(s) returned
✓ 18	23:34:46	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIM...	8 row(s) returned
✓ 19	23:36:26	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING S...	1 row(s) returned
✓ 20	23:37:13	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	2 row(s) returned
✓ 21	23:38:05	SELECT customer_id, COUNT(*) AS total_orders FROM orders GROUP BY customer_id HAVING COUNT...	3 row(s) returned
✓ 22	23:38:53	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	5 row(s) returned

Step 8 : BETWEEN + HAVING

```

100  -- Step 20: BETWEEN + HAVING
101  • SELECT customer_id,
102      MAX(purch_amt) AS max_purchase
103  FROM orders
104  GROUP BY customer_id
105  HAVING MAX(purch_amt) BETWEEN 2000 AND 6000;

```

customer_id	max_purchase
3002	5760.00
3009	2480.40
3007	2400.60

Result 19 x

Output

Action Output

#	Time	Action	Message
✓ 18	23:34:46	SELECT customer_id, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY customer_id LIMIT...	8 row(s) returned
✓ 19	23:36:26	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING S...	1 row(s) returned
✓ 20	23:37:13	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	2 row(s) returned
✓ 21	23:38:05	SELECT customer_id, COUNT(*) AS total_orders FROM orders GROUP BY customer_id HAVING COUNT...	3 row(s) returned
✓ 22	23:38:53	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	5 row(s) returned
✓ 23	23:39:42	SELECT customer_id, MAX(purch_amt) AS max_purchase FROM orders GROUP BY customer_id HAVIN...	3 row(s) returned

Step 9 : COUNT+HAVING

```

106  -- Step 21: COUNT + HAVING
107  • SELECT salesman_id,
108      COUNT(*) AS total_orders
109  FROM orders
110  GROUP BY salesman_id
111  HAVING COUNT(*) >= 2;

```

salesman_id	total_orders
5002	3
5001	4
5003	2

Result 20 x

Output

Action Output

#	Time	Action	Message
✓ 19	23:36:26	SELECT salesman_id, SUM(purch_amt) AS total_sales FROM orders GROUP BY salesman_id HAVING S...	1 row(s) returned
✓ 20	23:37:13	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	2 row(s) returned
✓ 21	23:38:05	SELECT customer_id, COUNT(*) AS total_orders FROM orders GROUP BY customer_id HAVING COUNT...	3 row(s) returned
✓ 22	23:38:53	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVIN...	5 row(s) returned
✓ 23	23:39:42	SELECT customer_id, MAX(purch_amt) AS max_purchase FROM orders GROUP BY customer_id HAVIN...	3 row(s) returned
✓ 24	23:44:56	SELECT salesman_id, COUNT(*) AS total_orders FROM orders GROUP BY salesman_id HAVING COUN...	3 row(s) returned

Step 10 : MAX+GROUP BY + HAVING

```

112 -- Step 22: MAX + GROUP BY + HAVING
113 • SELECT ord_date,
114       MAX(purch_amt) AS highest_purchase
115 FROM orders
116 GROUP BY ord_date
117 HAVING MAX(purch_amt) > 2000;

```

Result Grid  Filter Rows: Export:  Wrap Cell Content: 

	ord_date	highest_purchase
▶	2012-10-10	2480.40
	2012-07-27	2400.60
	2012-09-10	5760.00
	2012-04-25	3045.60

Result 21 x

Output

 Action Output

#	Time	Action	Message
✓ 20	23:37:13	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING COUNT(*) AS total_orders FROM orders GROUP BY customer_id HAVING COUNT(*) AS total_orders	2 row(s) returned
✓ 21	23:38:05	SELECT customer_id, COUNT(*) AS total_orders FROM orders GROUP BY customer_id HAVING COUNT(*) AS total_orders	3 row(s) returned
✓ 22	23:38:53	SELECT customer_id, SUM(purch_amt) AS total_purchase FROM orders GROUP BY customer_id HAVING COUNT(*) AS total_orders	5 row(s) returned
✓ 23	23:39:42	SELECT customer_id, MAX(purch_amt) AS max_purchase FROM orders GROUP BY customer_id HAVING COUNT(*) AS total_orders	3 row(s) returned
✓ 24	23:44:56	SELECT salesman_id, COUNT(*) AS total_orders FROM orders GROUP BY salesman_id HAVING COUNT(*) AS total_orders	3 row(s) returned
✓ 25	23:45:47	SELECT ord_date, MAX(purch_amt) AS highest_purchase FROM orders GROUP BY ord_date HAVING MAX(purch_amt) > 2000	4 row(s) returned

